HUJI · 769001 · DYNAMICS AND CODING IN THE BRAIN

Week 02

The Autapse: Fixed Points, Stability and Bistability

autapse · linear activation · fixed point · perturbation · stability · vector field · logistic function · bifurcation · bistability · memory

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SECTION 01

Where we are

Week 1 ended with the rate model of a network of N neurons [LMLF eq. 2.8-2.9]:

$$\tau \dot{r}_i = -r_i + g(h_i), \qquad h_i = \sum_{j=1}^{N} J_{ij} r_j + h_i^{\text{ext}}.$$

- r_i is the firing rate of neuron i; τ is its time constant ($\sim$10 ms).
- h_i is its total input: what it hears from the other neurons ($J_{ij} r_j$, synapse *from j to i*) plus external input h_i^{ext}.
- g is the activation function: it turns input into a rate.

THE PLAN FOR THIS WEEK

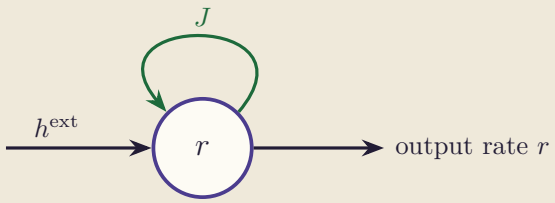
Before analysing big networks, study the smallest possible network: one neuron. If it talks to itself, it already has *feedback*, and feedback is what makes networks interesting. This week we use it to introduce two tools we will use all course long: **fixed points** and their **stability**. We do it twice: first with a linear activation function, where everything can be solved exactly, then with a saturating one (§10), where the same neuron becomes **bistable** and can store a memory.

SECTION 02

The autapse

DEF · AUTAPSE

An autapse is a synapse that a neuron makes onto itself: its axon connects back to its own dendrites. A neuron with an autapse is *the simplest possible network*: one neuron, one connection, of strength J [LMLF 2.3.1].



The autapse (after [LMLF Fig. 2.6]). The neuron receives an external input h^{ext}, and its own output r comes back to it through the synapse of strength J.

With a single neuron, the network equations lose their indices [LMLF eq. 2.11-2.12]:

AUTAPSE DYNAMICS

$$\tau \dot{r} = -r + g(h), \qquad h = Jr + h^{\text{ext}}.$$

DEF · EXCITATORY AUTAPSE

We focus on excitatory autapses, $J > 0$: the more the neuron fires, the more input it gives itself. This is positive feedback.

EXAMPLE · ONE INPUT VALUE, TWO SOURCES

$J = 0.5$, $h^{\text{ext}} = 1$, current rate $r = 4$: total input $h = 0.5 \cdot 4 + 1 = 3$. Two of the three units of input come from the neuron itself.

SECTION 03

The linear autapse

DEF · LINEAR ACTIVATION FUNCTION

The linear (identity) activation function is $g(h) = h$: the rate equals the input, with no threshold and no saturation [LMLF eq. 2.4].

PITFALL · WHY LINEAR IS A SIMPLIFICATION

A real neuron cannot fire at a negative rate, and it cannot fire faster than a few hundred Hz. The linear g ignores both limits. We use it because the resulting equation is linear, so we can solve it and understand it completely. It is a good approximation when the rate changes only a little.

With $g(h) = h$ the autapse equation becomes

$$\tau\dot{r} = -r + Jr + h^{\text{ext}} \qquad \Longrightarrow \qquad \boxed{\tau\dot{r} = -(1-J)\,r + h^{\text{ext}}}$$

TWO FORCES ON r

- $-r$: the leak. Left alone, the rate decays back to zero (Week 1).
- $+Jr$: the feedback. The neuron excites itself in proportion to its own rate.

The whole story of this week is the contest between leak and feedback: who wins depends on whether $J < 1$ or $J > 1$.

This is exactly the equation $\dot{n} = an + I$ from the Week 1 bonus hour, with $a = -(1-J)/\tau$ and constant input $I = h^{\text{ext}}/\tau$. We could solve it right away (we do so in §6). But first we introduce an approach that works *even when the equation cannot be solved*: look for the fixed points and ask whether they are stable.

SECTION 04

Fixed points

DEF · FIXED POINT

A fixed point r^* is a value of r where the dynamics stand still: $\dot{r} = 0$. If the system starts exactly at r^*, it stays there forever [LMLF 2.3.1].

Finding it. Set the right-hand side to zero:

$$0 = -(1-J)\,r^* + h^{\text{ext}} \qquad \Longrightarrow \qquad \boxed{r^* = \frac{h^{\text{ext}}}{1-J}} \qquad (J \neq 1).$$

EXAMPLE · FIXED POINTS FOR $h^{\text{ext}} = 1$

J	$r^* = 1/(1-J)$	COMMENT
0	1	no feedback: the rate just copies the input
0.5	2	feedback doubles the response
0.9	10	strong feedback: ten times the input
0.99	100	close to $J = 1$ the fixed point runs off
1.5	-2	negative! (see below)

FEEDBACK AMPLIFIES

For $0 < J < 1$, $r^* = h^{\text{ext}}/(1-J) > h^{\text{ext}}$: the autapse amplifies its input by the factor $1/(1-J)$. Why: the input raises r, which raises the feedback Jr, which raises r further, and so on. The extra contributions add up to $h^{\text{ext}}(1 + J + J^2 + \dots) = h^{\text{ext}}/(1-J)$, a geometric series that converges only if $J < 1$.

PITFALL · A FIXED POINT IS NOT AUTOMATICALLY WHERE THE SYSTEM GOES

For $J = 1.5$ the formula gives $r^* = -2$. It is a perfectly valid solution of $\dot{r} = 0$, but (as we will see) the system runs *away* from it. Also, a negative rate shows that the linear model has left the range where it describes a real neuron. Finding a fixed point is only half the job; the other half is its stability.

PITFALL · THE BORDER CASE $J = 1$

For $J = 1$ the leak and the feedback cancel exactly: $\tau\dot{r} = h^{\text{ext}}$. If $h^{\text{ext}} \neq 0$ there is *no* fixed point (r grows or shrinks steadily). If $h^{\text{ext}} = 0$ then $\dot{r} = 0$ everywhere: *every* value of r is a fixed point. The book returns to this special case later (§4.2); this week we assume $J \neq 1$.

SECTION 05

Stability

> 5.1 The idea

DEF · STABLE AND UNSTABLE FIXED POINTS

Start the system *near* r^* instead of exactly on it.

- If the dynamics bring r ever closer to r^*, the fixed point is stable.
- If the dynamics push r away from r^*, the fixed point is unstable.



stable: a ball in a valley unstable: a ball on a hilltop

Both balls are at rest (fixed points). A small nudge brings the left ball back but sends the right one rolling away.

WHY STABILITY MATTERS

In the real brain the rate is never exactly at r^*: there is always noise. An unstable fixed point is never seen in practice, because the tiniest disturbance grows. Only stable fixed points describe states the network actually sits in.

> 5.2 Perturbation analysis, step by step

To decide stability, follow a small deviation from the fixed point [LMLF eq. 2.14-2.15].

DEF · PERTURBATION

The perturbation is the distance from the fixed point: $\delta r(t) = r(t) - r^*$. Equivalently $r = r^* + \delta r$.

1. **Substitute** $r = r^* + \delta r$ into $\tau \dot{r} = -r + Jr + h^{\text{ext}}$:

$$\tau \frac{d}{dt}(r^* + \delta r) = -(r^* + \delta r) + J(r^* + \delta r) + h^{\text{ext}}.$$

2. r^* **is a number, not a function of time**, so $\frac{d}{dt} r^* = 0$ and the left side is $\tau\, \dot{\delta r}$.

3. **Regroup the right side** into the part with r^* and the part with δr:

$$\tau\, \dot{\delta r} = \underbrace{\big(-r^* + Jr^* + h^{\text{ext}}\big)}_{=0 \text{ (definition of } r^*)} + \big(-\delta r + J\delta r\big).$$

4. **The first bracket vanishes** because r^* is a fixed point. What remains is

$$\boxed{\tau\, \dot{\delta r} = -(1 - J)\, \delta r}$$

WHAT HAPPENED

The external input h^{ext} has disappeared. It only decides *where* the fixed point is; whether the fixed point is stable depends only on J.

> 5.3 Reading off stability from the sign

Case $J < 1$, so $1 - J > 0$:

- if $\delta r > 0$ (rate above the fixed point) then $\dot{\delta r} < 0$: it comes down;
- if $\delta r < 0$ (rate below) then $\dot{\delta r} > 0$: it goes up.

Either way the perturbation shrinks. The fixed point is stable.

Case $J > 1$, so $1 - J < 0$: now $\dot{\delta r}$ has the *same* sign as δr. Above the fixed point the rate rises further; below it, it falls further. The fixed point is unstable.

STABILITY OF THE LINEAR AUTAPSE

$$J < 1 \;\Rightarrow\; \text{stable} \qquad\qquad J > 1 \;\Rightarrow\; \text{unstable}$$

$J = 1$ is the border between two qualitatively different behaviours.

LEAK VS. FEEDBACK AGAIN

$J < 1$: when the rate goes up by δr, the leak pulls it down by δr but the feedback pushes it up only by $J\delta r < \delta r$. Leak wins, the deviation dies. $J > 1$: the feedback push $J\delta r$ beats the leak, the deviation feeds itself.

> 5.4 How fast? Solving for $\delta r(t)$

$\tau \dot{\delta r} = -(1 - J)\delta r$ is the Week 1 equation $\dot{n} = an$ with $a = -(1 - J)/\tau$, so [LMLF eq. 2.16]

PERTURBATION OVER TIME

$$\delta r(t) = \delta r(0)\, e^{-(1-J)t/\tau}.$$

- $J < 1$: exponential **decay** to 0, i.e. $r(t) \to r^*$.
- $J > 1$: exponential **growth**; r diverges to $+\infty$ or $-\infty$, depending on the sign of $\delta r(0)$.

EXAMPLE · HOW LONG A PERTURBATION LASTS, $\tau = 10$ MS

The perturbation shrinks (or grows) by a factor e every $\tau/|1-J|$:

J	$\tau/\lvert 1-J\rvert$	BEHAVIOUR
0	10 ms	decays (just the leak)
0.5	20 ms	decays, twice as slowly
0.9	100 ms	decays, ten times as slowly
1.5	20 ms	grows by e every 20 ms

The closer J is to 1, the slower everything happens: stronger feedback lets the neuron hold on to a deviation for much longer than its own τ.

SECTION 06

The full solution

Using the Week 1 result for $\dot{n} = an + I$ with constant input (or simply $r = r^* + \delta r$ with $\delta r(t)$ from above):

LINEAR AUTAPSE, SOLVED

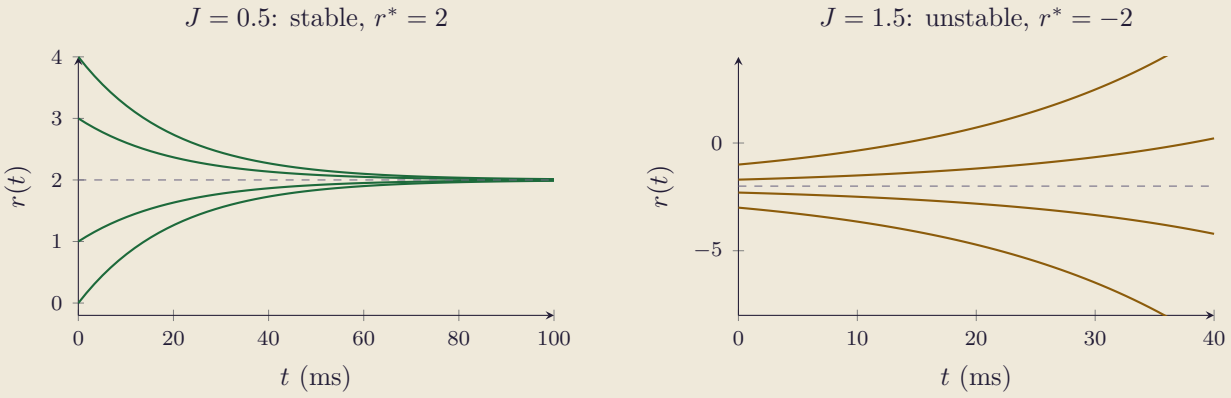
$$r(t) = r^* + \big(r(0) - r^*\big)\, e^{-(1-J)t/\tau}, \qquad r^* = \frac{h^{\text{ext}}}{1-J}.$$

Check. At $t = 0$: $r(0) = r^* + r(0) - r^* = r(0)$ ✓. Differentiate: $\tau\dot{r} = -(1-J)(r(0)-r^*)e^{-(1-J)t/\tau} = -(1-J)(r - r^*) = -(1-J)r + (1-J)r^* = -(1-J)r + h^{\text{ext}}$. ✓

EXAMPLE · STARTING FROM SILENCE, $r(0) = 0$, $h^{\text{ext}} = 1$, $\tau = 10$ MS

t (ms)	0	10	20	40	60	100
$J = 0.5$: $2(1 - e^{-t/20})$	0	0.79	1.26	1.73	1.90	1.99
$J = 0.9$: $10(1 - e^{-t/100})$	0	0.95	1.81	3.30	4.51	6.32
$J = 1.5$: $-2 + 2e^{t/20}$	0	1.30	3.44	12.8	38.2	295

$J = 0.5$ settles at 2 within ∼60 ms. $J = 0.9$ heads for 10 but needs hundreds of ms. $J = 1.5$ starts *above* its fixed point -2, so it is pushed further and further up, without limit.



$r(t)$ from four starting points ($\tau = 10\,\text{ms}$, $h^{\text{ext}} = 1$). Left: every trajectory converges to the dashed fixed point. Right: every trajectory that does not start exactly on the fixed point is driven away from it.

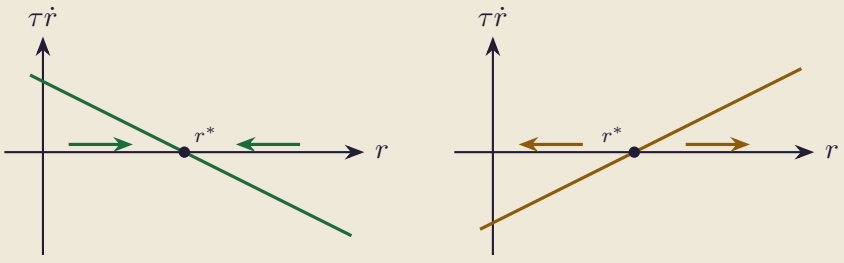
SECTION 07

The geometric view: the vector field

We can see stability *without solving anything*, by plotting $\dot{r}$ as a function of r (the idea from the Week 1 bonus hour).

DEF · VECTOR FIELD ON A LINE

The equation $\tau\dot{r} = F(r)$ assigns a velocity $\dot{r}$ to every value of r: a vector field on the line [LMLF 2.3.1]. Draw $\dot{r}$ against r and put arrows on the r-axis: right where $\dot{r} > 0$, left where $\dot{r} < 0$. The fixed points are where the graph crosses zero.



$J < 1$: slope $-(1-J) < 0$
arrows point *in*: stable

$J > 1$: slope $J - 1 > 0$
arrows point *out*: unstable

$\tau\dot{r} = -(1-J)r + h^{\text{ext}}$ is a straight line in r that crosses zero at r^*. Its slope decides everything.

STABILITY FROM THE SLOPE

A fixed point is stable if the graph of $\dot{r}$ against r crosses zero going **down** (negative slope), and unstable if it crosses going **up** (positive slope). For the linear autapse the slope is $-(1-J)/\tau$: negative exactly when $J < 1$.

WHY THIS IS POWERFUL

This picture needs only the *sign* of $\dot{r}$ around r^*. It works for equations we cannot solve, which will be the usual case once g is not linear.

SECTION 08

Connection to Week 1: it is the eigenvalue

In the TA session of Week 1 we saw that $\dot{\vec{x}} = M\vec{x}$ converges iff all eigenvalues of M are negative. The perturbation equation

$$\dot{\delta r} = \underbrace{-\frac{1-J}{\tau}}_{\text{a } 1\times 1 \text{ “matrix”}} \delta r$$

is the one-dimensional case: its only eigenvalue is $-(1-J)/\tau$, negative exactly when $J < 1$. Stability of a fixed point = the sign of the eigenvalue of the perturbation dynamics. For networks of many neurons the same recipe gives a matrix, and Week 1's eigenvalue tools take over.

SECTION 09

A limitation of the linear model

PITFALL · DIVERGENCE IS NOT BIOLOGY

For $J > 1$ the linear autapse fires at a rate that grows without bound (or goes to $-\infty$). Real neurons *saturate*: their rate cannot exceed a few hundred Hz, and it cannot go below zero. This is a limitation of the linear g, not of the autapse itself. It is the reason the book next replaces $g(h) = h$ by a saturating activation function [LMLF 2.3.1].

SECTION 10

The non-linear autapse: a saturating activation function

The rest of this week keeps the *same* autapse, $\tau\dot{r} = -r + g(Jr + h^{\text{ext}})$, and changes only g. The result is surprising: a single neuron with strong feedback no longer explodes. Instead it gets two stable states, and it can use them to remember a bit of information.

Two misreadings to clear up first:

- **The non-linearity does not make the analysis hopeless.** We can no longer write down $r(t)$, but fixed points and their stability can still be found, with a picture.
- **Stability is no longer a property of the whole system.** For the linear autapse, J alone decided stability. Now each fixed point has *its own* stability, and one neuron can have a stable

and an unstable fixed point side by side.

> 10.1 The logistic function

DEF · LOGISTIC ACTIVATION FUNCTION

The logistic function [LMLF eq. 2.2] is

$$g(x) = \frac{1}{1 + e^{-\beta(x-\theta)}}.$$

It is sigmoidal (S-shaped): it increases monotonically, it is bounded below by 0 (no negative rates) and above by 1 (saturation). The rate r is therefore *normalised*: $r = 1$ means "firing at the neuron's maximal rate", not 1 Hz [LMLF 2.2].

It has two parameters:

DEF · THRESHOLD θ AND GAIN β

- The threshold θ is the input at which the rate is exactly half its maximum: $g(\theta) = 1/(1+e^0) = 0.5$. Changing θ slides the curve left or right.
- The gain β sets how steep the curve is. Its maximal slope, reached at $x = \theta$, is $\beta/4$ (derived below).

EXAMPLE · THE LOGISTIC FUNCTION WITH $\theta = 0$, $\beta = 20$ (THE BOOK'S VALUES)

x	-0.3	-0.2	-0.1	-0.05	0	0.05	0.1	0.2	0.3
$g(x)$	0.002	0.018	0.119	0.269	0.5	0.731	0.881	0.982	0.998
$g'(x)$	0.05	0.35	2.1	3.9	5	3.9	2.1	0.35	0.05

Almost all of the change happens for x between -0.2 and 0.2. Outside that window the neuron is either nearly silent or nearly saturated, and the slope is close to 0.

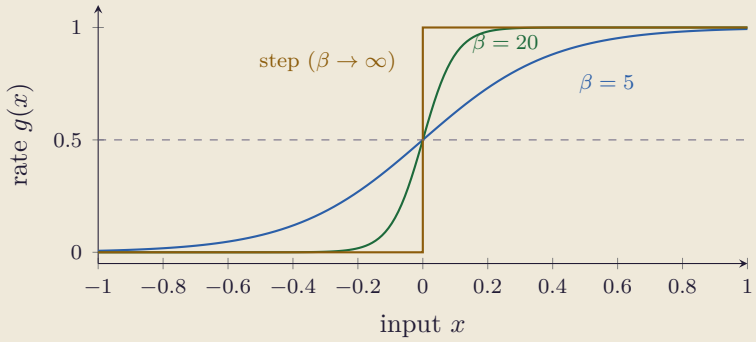
The slope of the logistic function, step by step (we need it for stability). Write $u = e^{-\beta(x-\theta)}$, so $g = 1/(1+u)$ and $du/dx = -\beta u$:

$$g'(x) = -\frac{1}{(1+u)^2} \cdot (-\beta u) = \beta \, \frac{1}{1+u} \cdot \frac{u}{1+u} = \beta \, g(x)(1 - g(x)).$$

The product $g(1-g)$ is largest when $g = \frac{1}{2}$, i.e. at $x = \theta$, where it equals $\frac{1}{4}$. So

SLOPE OF THE LOGISTIC

$$g'(x) = \beta \, g(x) \, (1 - g(x)), \qquad \max_x g'(x) = g'(\theta) = \frac{\beta}{4}.$$



The logistic function with $\theta = 0$ (after [LMLF Fig. 2.2a]). All curves pass through $(\theta, 0.5)$. A larger gain β gives a steeper curve. In the limit $\beta \to \infty$ it becomes a step: 0 below threshold, 1 above.

> 10.2 The fixed-point equation

Setting $\dot{r} = 0$ in $\tau\dot{r} = -r + g(Jr + h^{\text{ext}})$ gives [LMLF eq. 2.17]

FIXED POINTS OF THE NON-LINEAR AUTAPSE

$$r^* = g(Jr^* + h^{\text{ext}}).$$

PITFALL · YOU CANNOT SOLVE THIS FOR r^* WITH ALGEBRA

r^* appears on both sides, once inside an exponential. There is no formula like $r^* = h^{\text{ext}}/(1 - J)$ any more. Do not try to isolate r^*. Instead, read the equation as a question about two curves.

The geometric method, step by step:

1. **Left side**: plot $y = r$ against r. This is the identity line (slope 1).
2. **Right side**: plot $y = g(Jr + h^{\text{ext}})$ against r. As a function of r this is *again a logistic function*, with new parameters. Substitute $x = Jr + h^{\text{ext}}$:

   $$\beta(Jr + h^{\text{ext}} - \theta) = (\beta J)\Big(r - \frac{\theta - h^{\text{ext}}}{J}\Big),$$

   so it has effective gain βJ and effective threshold $(\theta - h^{\text{ext}})/J$ [LMLF 2.3.1].
3. **Every intersection** of the two curves satisfies $r = g(Jr + h^{\text{ext}})$, so it is a fixed point. Counting intersections counts fixed points.

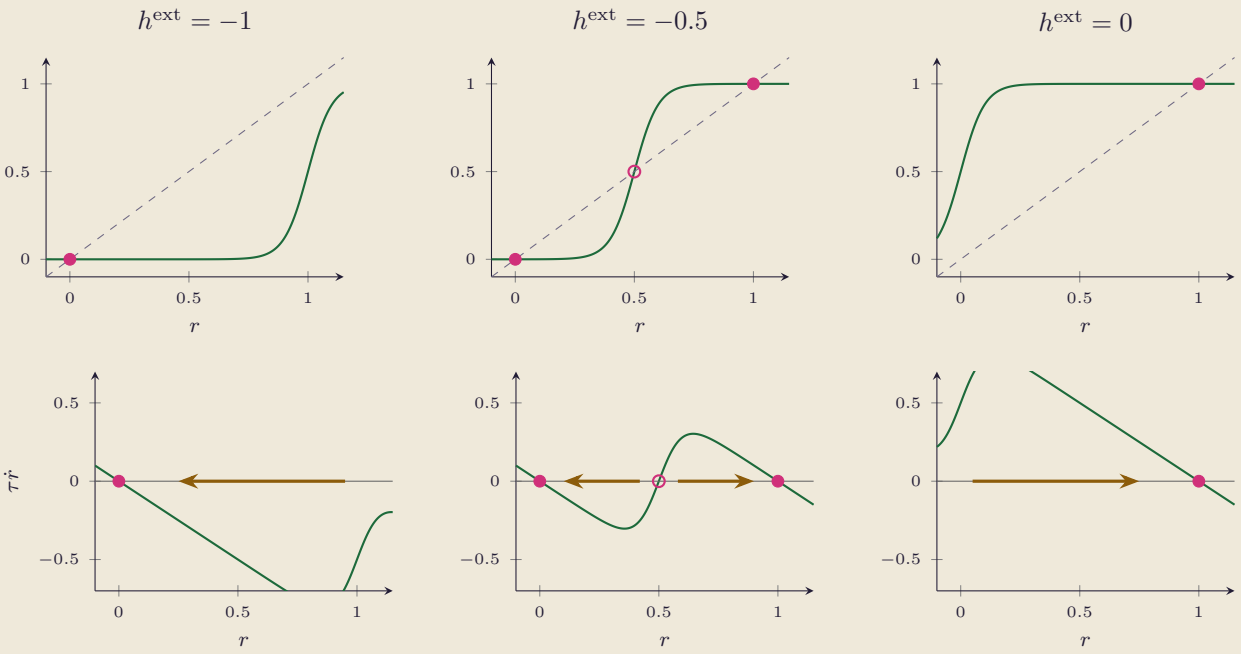
WHAT h^{ext} DOES IN THE PICTURE

h^{ext} appears only in the effective threshold $(\theta - h^{\text{ext}})/J$. Raising h^{ext} lowers the threshold, so it slides the S-curve to the left, along the r axis. The neuron then reaches half its maximal rate already at a smaller r, because the external input supplies part of the drive.

PITFALL · LEFT, NOT RIGHT

The book writes that increasing h^{ext} "results in a shift of the logistic function to the right". As a function of r, the curve moves to the *left*: its midpoint $(\theta - h^{\text{ext}})/J$ decreases when h^{ext} grows. Check with the book's own numbers ($\theta = 0$, $J = 1$): the midpoint is at $r = 1$ for

$h^{\text{ext}} = -1$, at $r = 0.5$ for $h^{\text{ext}} = -0.5$ and at $r = 0$ for $h^{\text{ext}} = 0$. The figure below (and [LMLF Fig. 2.7]) shows exactly this leftward slide.



Redrawn after [LMLF Fig. 2.7]; $\theta = 0$, $\beta = 20$, $J = 1$. **Top**: the S-curve $g(r + h^{\text{ext}})$ (green) and the identity line (dashed); the intersections are the fixed points. **Bottom**: the vector field $\tau\dot{r} = -r + g(r + h^{\text{ext}})$ with the direction of motion on the r axis (arrows). Filled dots: stable fixed points; open dot: unstable. Left: one low-activity fixed point. Middle: three fixed points. Right: one high-activity fixed point.

EXAMPLE · FINDING THE FIXED POINTS FOR $h^{\text{ext}} = -0.5$ BY HAND

The equation is $r^* = g(r^* - 0.5)$ with $g(x) = 1/(1 + e^{-20x})$.

- **Middle**: try $r^* = 0.5$. Then $g(0) = 0.5$ ✓. It is a fixed point exactly.
- **Low**: try $r^* \approx 0$. Then $g(-0.5) = 1/(1 + e^{10}) \approx 4.5 \times 10^{-5}$. So $r^* \approx 0.000045$: the neuron is practically silent.
- **High**: try $r^* \approx 1$. Then $g(0.5) = 1/(1 + e^{-10}) \approx 1 - 4.5 \times 10^{-5}$. So $r^* \approx 0.99995$: practically saturated.

Each guess is then improved by plugging the result back in, but the change is in the fifth decimal place.

> 10.3 Stability from the vector field

We read stability off the bottom row, exactly as in the linear case: arrows pointing *in* mean stable, arrows pointing *out* mean unstable.

EXAMPLE · THE SIGN OF $\tau\dot{r}$ FOR $h^{\text{ext}} = -0.5$

r	0	0.1	0.25	0.4	**0.5**	0.6	0.75	0.9	1
$g(r-0.5)$	0.000	0.000	0.007	0.119	0.5	0.881	0.993	1.000	1.000
$\tau\dot{r}$	≈ 0	-0.10	-0.24	-0.28	**0**	$+0.28$	$+0.24$	$+0.10$	≈ 0
motion	•	$\leftarrow$	$\leftarrow$	$\leftarrow$	$\circ$	$\rightarrow$	$\rightarrow$	$\rightarrow$	•

Between 0 and 0.5 the rate falls, towards 0. Between 0.5 and 1 it rises, towards 1. So 0 and 1 attract, and 0.5 repels: it is the watershed that decides which way the neuron goes.

For a single fixed point (left and right panels) the arrows point towards r^* from *every* starting value. For three fixed points, the outer ones attract only the starting values on their own side of 0.5. The book separates these two situations [LMLF 2.3.1]:

DEF · GLOBAL AND LOCAL STABILITY

- A fixed point is globally stable if the dynamics reach it from every starting value $r \neq r^*$. Example: the only fixed point for $h^{\text{ext}} = -1$.
- It is locally stable if the dynamics reach it from every starting value close enough to it, but not necessarily from far away. Example: $r^* \approx 0$ for $h^{\text{ext}} = -0.5$. Starting at $r = 0.3$ leads there, starting at $r = 0.7$ does not.

Every globally stable fixed point is locally stable; the converse is false.

> 10.4 Stability by linearisation

The picture is convincing, but we also want a number, as in the linear case. The trick is to make the equation linear near the fixed point.

DEF · FIRST-ORDER TAYLOR EXPANSION

Near a point a, a smooth function is approximately a straight line: $f(a+\varepsilon) \approx f(a) + f'(a)\,\varepsilon$ for small ε. **Example**: $g(x) = 1/(1+e^{-20x})$ at $a = 0.1$, with $\varepsilon = 0.01$. Exact: $g(0.11) = 0.9002$. Approximation: $g(0.1) + g'(0.1) \cdot 0.01 = 0.8808 + 2.1 \cdot 0.01 = 0.9018$. Close, because ε is small.

Follow the same recipe as for the linear autapse [LMLF eq. 2.18-2.22]:

1. **Substitute** $r = r^* + \delta r$:

$$\tau \frac{d}{dt}(r^* + \delta r) = -(r^* + \delta r) + g\big(J(r^* + \delta r) + h^{\text{ext}}\big).$$

2. **Name the input at the fixed point** $h^* = Jr^* + h^{\text{ext}}$. The argument of g is then $h^* + J\delta r$.

3. **Taylor-expand** g around h^* with the small step $\varepsilon = J\delta r$:

$$g(h^* + J\delta r) \approx g(h^*) + g'(h^*)\, J\delta r.$$

4. **Regroup** (and use $\frac{d}{dt} r^* = 0$):

$$\tau\,\dot{\delta r} = \underbrace{\big(-r^* + g(h^*)\big)}_{=0 \text{ (fixed point)}} - \delta r + J g'(h^*)\,\delta r.$$

5. **Result**, with the number $\lambda = Jg'(h^*) = Jg'(Jr^* + h^{\text{ext}})$:

$$\boxed{\tau\,\dot{\delta r} = -(1-\lambda)\,\delta r} \qquad \Longrightarrow \qquad \delta r(t) = \delta r(0)\, e^{-(1-\lambda)t/\tau}.$$

LOCAL STABILITY OF A FIXED POINT OF THE AUTAPSE

$$\lambda = J\,g'(Jr^* + h^{\text{ext}}): \qquad \lambda < 1 \;\Rightarrow\; \text{stable}, \qquad \lambda > 1 \;\Rightarrow\; \text{unstable}.$$

THE SAME EQUATION AS BEFORE, WITH J REPLACED BY λ

Compare with $\tau\dot{\delta r} = -(1-J)\delta r$ for the linear autapse. Near a fixed point, the non-linear neuron behaves like a *linear* autapse whose feedback strength is the effective feedback $\lambda = Jg'(h^*)$: the synaptic weight times how sensitive the neuron is at its current operating point. For $g(h) = h$, $g' = 1$ and $\lambda = J$: we recover the old result.

EXAMPLE · λ AT THE THREE FIXED POINTS FOR $h^{\text{ext}} = -0.5$ ($J = 1$, $\beta = 20$, $\tau = 10\,\text{MS}$)

Use $g' = \beta g(1-g)$ and $g(h^*) = r^*$ at a fixed point, so $\lambda = J\beta\, r^*(1-r^*)$:

r^*	$\lambda = 20\, r^*(1-r^*)$	VERDICT	TIME SCALE $\tau/\lvert 1-\lambda\rvert$
0.000045	0.0009	stable	≈ 10 ms decay
0.5	5	unstable	2.5 ms growth
0.99995	0.0009	stable	≈ 10 ms decay

At the outer fixed points the neuron is silent or saturated, so the curve is flat ($g' \approx 0$): feedback does nothing and only the leak acts. At 0.5 the curve is steepest: a deviation of 0.01 grows by a factor e every 2.5 ms.

The geometric version. $\lambda = Jg'(Jr^* + h^{\text{ext}}) = \frac{d}{dr}\, g(Jr + h^{\text{ext}})\big|_{r^*}$ is the slope of the S-curve at the intersection. The identity line has slope 1. So:

STABILITY AT A GLANCE

In the top-row picture, a fixed point is **stable** if the S-curve crosses the identity line **less steeply** than it (slope < 1), and **unstable** if it crosses **more steeply** (slope > 1).

Look again at the figure: the single crossings (left, right) and the two outer crossings (middle) are flat; the middle crossing is steep.

> 10.5 Bifurcations

As h^{ext} increases smoothly from -1 to 0, the number of fixed points jumps: $1 \to 3 \to 1$. This kind of event has a name [LMLF 2.3.1].

DEF · BIFURCATION

A bifurcation is a *qualitative* change in the dynamics (fixed points appear or disappear, or a fixed point changes its stability) caused by a *smooth* change of a parameter. The parameter values where it happens are the bifurcation points.

DEF · SADDLE-NODE BIFURCATION

In a saddle-node bifurcation a stable and an unstable fixed point move towards each other, merge into one, and disappear (or, run backwards, are born together out of nothing). Both transitions of the autapse are of this type. At the moment of merging, the S-curve just touches the identity line: the two curves are tangent.

EXAMPLE · COMPUTING THE TWO BIFURCATION POINTS ($\theta = 0$, $\beta = 20$, $J = 1$)

At a bifurcation point the S-curve is tangent to the identity line, so two conditions hold at once:

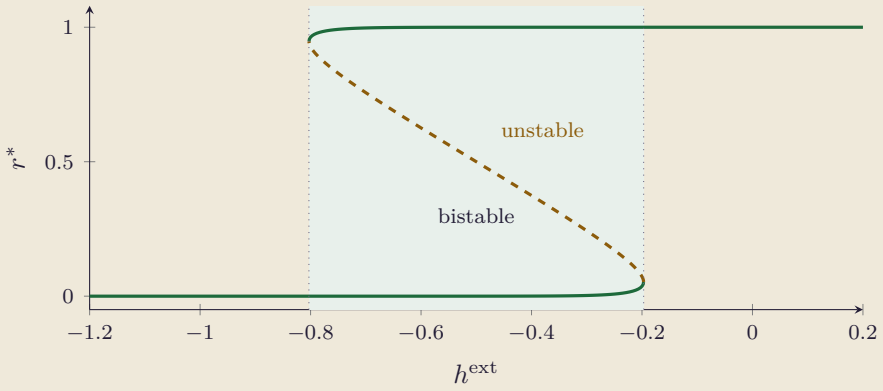
1. it is a fixed point: $r^* = g(h^*)$, with $h^* = r^* + h^{\text{ext}}$;
2. the slopes are equal: $\lambda = J\beta\, g(1-g) = 1$, i.e. $20\, r^*(1-r^*) = 1$.

Step 1. Condition 2 is a quadratic: $r^{*2} - r^* + 0.05 = 0$, so $r^* = \frac{1}{2}(1 \pm \sqrt{0.8})$, i.e. $r^* = 0.947$ or $r^* = 0.053$.
Step 2. Invert the logistic to find the input: $g(h^*) = r^*$ gives $h^* = \frac{1}{\beta} \ln \frac{r^*}{1-r^*}$. For $r^* = 0.947$: $h^* = \frac{1}{20} \ln 17.9 = 0.144$. For $r^* = 0.053$: $h^* = -0.144$.
Step 3. $h^{\text{ext}} = h^* - Jr^*$: $0.144 - 0.947 = \mathbf{-0.803}$ and $-0.144 - 0.053 = \mathbf{-0.197}$.
So the autapse has three fixed points exactly for $-0.803 < h^{\text{ext}} < -0.197$. The book's examples $h^{\text{ext}} = -1$ (below), -0.5 (inside), 0 (above) fall where they should.

The whole story fits in one picture: plot every fixed point against h^{ext}.

DEF · BIFURCATION DIAGRAM

A bifurcation diagram plots the fixed points (vertical axis) against a parameter (horizontal axis). Stable fixed points are drawn solid, unstable ones dashed.



Bifurcation diagram, redrawn after [LMLF Fig. 2.8] ($\theta = 0$, $\beta = 20$, $J = 1$). Solid: stable fixed points; dashed: unstable. In the shaded band, $-0.803 < h^{\text{ext}} < -0.197$, there are three fixed points. At each edge of the band the dashed branch merges with a solid one: a saddle-node bifurcation.

> 10.6 Attractors and bistability

DEF · STATE SPACE AND ATTRACTOR

The state space is the set of all possible states of the system; for the autapse it is just the r axis. An attractor is a region of state space to which all nearby trajectories converge [LMLF 2.3.1]. A stable fixed point is the simplest attractor; others appear later in the course.

DEF · BISTABILITY

A system is bistable if it has two stable fixed points (two attractors). Between them there is always an unstable fixed point, which separates the starting values that go to one from those that go to the other. The autapse is bistable for $-0.803 < h^{\text{ext}} < -0.197$ (with $\theta = 0$, $\beta = 20$, $J = 1$).

EXAMPLE · WHERE DOES THE NEURON END UP? ($h^{\text{ext}} = -0.5$, $\tau = 10\,\text{MS}$)

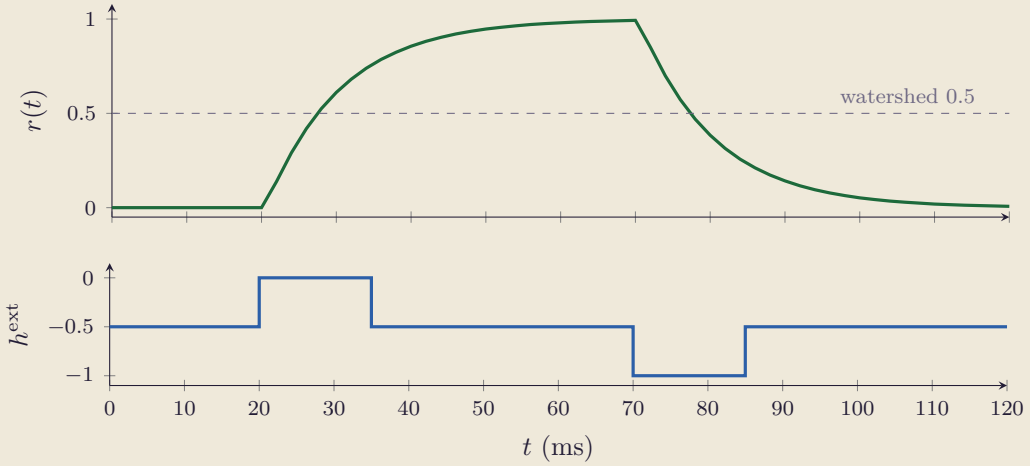
t (ms)	0	10	20	30	50	70	100
$r(0) = 0.40$	0.40	0.157	0.058	0.021	0.003	0.000	0.000
$r(0) = 0.45$	0.45	0.194	0.072	0.026	0.004	0.001	0.000
$r(0) = 0.55$	0.55	0.806	0.928	0.974	0.996	0.999	1.000
$r(0) = 0.60$	0.60	0.843	0.942	0.979	0.997	1.000	1.000

(Numerical solutions.) Starting values only 0.1 apart end in opposite states. The final state is decided entirely by which side of 0.5 the neuron starts on.

> 10.7 Memory: switching with a brief input

Why care about two stable states? Because a bistable neuron can store one bit [LMLF 2.3.1]. Left alone in the bistable range, it stays forever in whichever state it is in. To write a bit, change h^{ext} *briefly*:

1. The neuron rests in the low state, $h^{\text{ext}} = -0.5$.
2. A brief pulse raises h^{ext} to 0, *above* the bifurcation point -0.197. Now the low fixed point no longer exists; the only attractor is the high one, and r starts climbing.
3. The pulse ends: h^{ext} returns to -0.5. The low state exists again, but if r has already passed the watershed 0.5, it keeps going to the high state.
4. The neuron stays high indefinitely, although the input that put it there is long gone. That is memory.
5. To erase, pulse h^{ext} *below* -0.803 (here to -1): the high state disappears and the neuron falls back to the low one, where it stays.



State transitions, simulated after [LMLF Fig. 2.9] ($\theta = 0$, $\beta = 20$, $J = 1$, $\tau = 10\,\text{ms}$). A 15 ms pulse to $h^{\text{ext}} = 0$ switches the neuron on (it has reached $0.76 > 0.5$ when the pulse ends); a 15 ms pulse to $h^{\text{ext}} = -1$ switches it off (it has fallen to $0.23 < 0.5$). In between, at the same $h^{\text{ext}} = -0.5$, the neuron is high or low depending only on its history.

THE STATE DEPENDS ON HISTORY

At $t = 10\,\text{ms}$ and at $t = 60\,\text{ms}$ the input is the same, $h^{\text{ext}} = -0.5$, but the rate is 0 in one case and 0.98 in the other. The difference is what happened before. This dependence on past input (often called hysteresis) is what lets a single neuron with feedback hold information; the book develops it into a model of long-term memory in Ch. 5.

PITFALL · THE PULSE MUST BE LONG ENOUGH

Raising h^{ext} is not enough by itself; the rate must cross the watershed 0.5 *before* the pulse ends. With the same parameters, a 5 ms pulse to $h^{\text{ext}} = 0$ brings r only to 0.36. When h^{ext} returns to -0.5, the rate is on the low side of 0.5 and decays back to 0: nothing is stored.

> 10.8 The two ingredients: positive feedback and saturation

What makes bistability possible? We need an S-curve that can cross the identity line three times. Two things are required [LMLF 2.3.1]:

- **Positive feedback** that is strong enough: in the middle crossing the S-curve must be steeper than 1. Its maximal slope is $(\beta J)/4$ (the effective gain over 4), so we need

$$\boxed{\frac{\beta J}{4} > 1}.$$

  If this fails, the S-curve is flatter than the identity line everywhere and crosses it only once, whatever h^{ext} is. If it holds, a suitable h^{ext} produces three crossings.

- **Saturation**: the curve must bend back below the identity line at high r (and stay above it at low r). That is what the bounded logistic function provides, and what the linear g lacked: there, $J > 1$ meant an unstable fixed point and divergence, with no second stable state to land in.

EXAMPLE · HOW MUCH FEEDBACK IS ENOUGH? ($\beta = 20$)

$\beta J/4 > 1 \iff J > 4/\beta = 0.2$. With the book's $J = 1$: $\beta J/4 = 5$, comfortably bistable for some range of h^{ext}. With $J = 0.1$: $\beta J/4 = 0.5$, one fixed point for every h^{ext}, no memory.

PITFALL · THREE INTERSECTIONS, NOT TWO

The book says that the bistable regime has "two intersection points". It has *three* intersections (fixed points), of which *two* are stable. The unstable middle one is always there, between the two attractors.

> 10.9 A technical aside: the case $\lambda = 1$

The linearisation decides stability whenever $\lambda \neq 1$. When $\lambda = 1$ exactly, it gives $\tau\dot{\delta r} = 0$, which says nothing. This happens precisely at the bifurcation points, where the S-curve is tangent to the identity line.

Step 1: expand one order further [LMLF eq. 2.23–2.25]. A second-order Taylor expansion adds the curvature term:

$$g(h^* + J\delta r) \approx g(h^*) + g'(h^*)J\delta r + \tfrac{1}{2}g''(h^*)J^2\delta r^2.$$

Step 2: substitute. The constant term vanishes (fixed point) and the linear term vanishes ($\lambda = 1$), leaving

$$\tau\,\dot{\delta r} = \tfrac{1}{2}J^2 g''(h^*)\,\delta r^2.$$

Step 3: read the sign. $\delta r^2 > 0$ on *both* sides of r^*, so $\dot{\delta r}$ has the same sign (that of g'') on both sides. The flow goes towards r^* on one side and away on the other.

DEF · HALF-STABLE FIXED POINT

A fixed point is half-stable if the dynamics approach it from one side and leave it on the other.

Step 4: solve (the equation is separable): $\dfrac{d\delta r}{\delta r^2} = c\,dt$ with $c = \frac{J^2 g''(h^*)}{2\tau}$, so $-\frac{1}{\delta r} = ct + \text{const}$ and

$$\delta r(t) = \frac{1}{1/\delta r(0) - c\,t}.$$

This is hyperbolic, not exponential: much slower near the fixed point.

EXAMPLE · AT THE BIFURCATION POINT $h^{\text{ext}} = -0.803$ ($\beta = 20$, $J = 1$, $\tau = 10\,\text{MS}$)

Here $r^* = 0.947$, and $g'' = \beta^2 g(1-g)(1-2g) = 400 \cdot 0.947 \cdot 0.053 \cdot (-0.894) = -17.9$, so $c = -17.9/20 = -0.89$ per ms.

- From above, $\delta r(0) = +0.01$: $\delta r(t) = 1/(100 + 0.89t)$. After 100 ms it is 0.0053: only halved. (An exponential decay with $\tau = 10$ ms would have shrunk it $e^{10} \approx 22\,000$-fold.)
- From below, $\delta r(0) = -0.01$: $\delta r(t) = 1/(-100 + 0.89t)$ grows in size and the approximation breaks down after about 100 ms; the neuron leaves towards the low state.

PITFALL · LINEARISATION TELLS YOU ONLY ABOUT THE NEIGHBOURHOOD

Both the first- and second-order expansions are valid only while δr is small. They decide *local* stability. Whether a fixed point is reached from far away (global stability) needs the full picture, such as the vector field [LMLF 2.3.1]. And $\lambda = 1$ exactly is a knife-edge: the slightest change of h^{ext} makes $\lambda \neq 1$, and the exponential behaviour takes over again.

SECTION 11

Practice problems with full solutions

EXAMPLE · P1: FIXED POINT AND STABILITY

$\tau\dot{r} = -r + 0.8r + 3$. Find r^*, decide stability, and give $r(t)$ for $r(0) = 0$, $\tau = 10\,\text{ms}$.

Solution. $J = 0.8$, $h^{\text{ext}} = 3$: $r^* = 3/0.2 = 15$. $J < 1$, so stable. $r(t) = 15(1 - e^{-0.2t/10}) = 15(1 - e^{-t/50})$: the time scale is 50 ms, five times τ.

EXAMPLE · P2: WHICH J GIVES A TARGET RATE?

With $h^{\text{ext}} = 2$, what J makes the stable fixed point $r^* = 8$?

Solution. $8 = 2/(1 - J) \Rightarrow 1 - J = 0.25 \Rightarrow J = 0.75$. Since $0.75 < 1$ it is stable. ✓

EXAMPLE · P3: PERTURBATION BY HAND

$J = 1.2$, $h^{\text{ext}} = -0.4$, $\tau = 10\,\text{ms}$. Find r^*. The rate starts at $r(0) = 2.1$; what is it after 50 ms?

Solution. $r^* = -0.4/(1 - 1.2) = -0.4/(-0.2) = 2$. $J > 1$: unstable. $\delta r(0) = 0.1$, and $\delta r(t) = 0.1\, e^{0.2t/10} = 0.1\, e^{t/50}$. At $t = 50$: $\delta r = 0.1e \approx 0.27$, so $r \approx 2.27$. The rate drifts away from 2, slowly at first and then faster and faster.

EXAMPLE · P4: VECTOR FIELD

Sketch $\tau\dot{r}$ against r for $\tau\dot{r} = -0.5r + 1$ and use it to say what happens from $r(0) = 5$.

Solution. A line with slope -0.5 crossing zero at $r^* = 2$. At $r = 5$: $\tau\dot{r} = -1.5 < 0$, so r decreases, and it keeps decreasing until it reaches 2. The arrows point towards 2 from both sides: stable.

EXAMPLE · P5: THE BORDER

What happens for $J = 1$, $h^{\text{ext}} = 0.5$, $\tau = 10\,\text{ms}$, $r(0) = 0$?

Solution. $\tau\dot{r} = 0.5$, so $\dot{r} = 0.05$ per ms: $r(t) = 0.05t$, a straight line. There is no fixed point; the rate grows steadily, not exponentially.

The non-linear autapse

EXAMPLE · P6: READING A LOGISTIC FUNCTION

$g(x) = 1/(1 + e^{-10(x-0.2)})$. Find $g(0.2)$, $g(0.4)$, $g(0)$ and the maximal slope.

Solution. $\theta = 0.2$, $\beta = 10$. $g(0.2) = 0.5$ (the threshold). $g(0.4) = 1/(1 + e^{-2}) = 0.881$. $g(0) = 1/(1 + e^{2}) = 0.119$ (symmetric about the threshold). Maximal slope $\beta/4 = 2.5$, at $x = 0.2$.

EXAMPLE · P7: EFFECTIVE THRESHOLD

$\theta = 0$, $J = 2$. Where along the r axis is the midpoint of the S-curve $g(Jr + h^{\text{ext}})$ for $h^{\text{ext}} = -1$? For $h^{\text{ext}} = -0.4$? Which way did it move?

Solution. Midpoint $(\theta - h^{\text{ext}})/J$: $(0 + 1)/2 = 0.5$, and $(0 + 0.4)/2 = 0.2$. Raising h^{ext} moved it *left*, from 0.5 to 0.2.

EXAMPLE · P8: STABILITY BY λ

$\theta = 0$, $\beta = 20$, $J = 1$, $h^{\text{ext}} = -0.3$. One fixed point is $r^* = 0.243$; check that it is one, and decide its stability. What about $r^* \approx 0.0026$?

Solution. $g(0.243-0.3) = g(-0.057) = 1/(1+e^{1.14}) = 0.242 \approx 0.243$ ✓. $\lambda = 20 \cdot 0.243 \cdot 0.757 = 3.68 > 1$: unstable. For $r^* = 0.0026$: $\lambda = 20 \cdot 0.0026 \cdot 0.9974 = 0.052 < 1$: stable. (The third fixed point, $r^* \approx 1$, is also stable: $h^{\text{ext}} = -0.3$ is inside the bistable band.)

EXAMPLE · P9: CAN IT BE BISTABLE AT ALL?

(a) $\beta = 2$, $J = 1$. (b) $\beta = 20$, $J = 0.15$. (c) $\beta = 8$, $J = 1$. For which can a suitable h^{ext} give two stable states?

Solution. Test $\beta J/4 > 1$: (a) 0.5, no. (b) 0.75, no. (c) 2, yes. In (a) and (b) the S-curve is everywhere flatter than the identity line, so it crosses it once whatever h^{ext} is.

EXAMPLE · P10: LINEAR AS A SPECIAL CASE

Show that the stability rule $\lambda < 1$ reproduces the linear autapse result.

Solution. $g(h) = h$ has $g' = 1$ everywhere, so $\lambda = J \cdot 1 = J$. The rule becomes $J < 1$: stable, $J > 1$: unstable, exactly as in the first half of the week.

EXAMPLE · P11: WRITING A BIT

With $\theta = 0$, $\beta = 20$, $J = 1$, $\tau = 10\,\text{ms}$, the neuron rests at the low state with $h^{\text{ext}} = -0.5$. You can pulse h^{ext} to 0. A 5 ms pulse brings r to 0.36, a 15 ms pulse to 0.76. Which pulse stores a "1"? Why does setting $h^{\text{ext}} = -0.3$ forever not count as memory?

Solution. Only the 15 ms pulse: 0.76 is above the unstable fixed point 0.5 of $h^{\text{ext}} = -0.5$, so the rate continues to the high state; 0.36 is below it and decays back. Memory means the state persists *after* the input returns to its baseline; a permanent change of h^{ext} is just a different input, not a stored bit.

EXAMPLE · P12: HALF-STABLE

At the bifurcation point $h^{\text{ext}} = -0.197$ ($\beta = 20$, $J = 1$), $r^* = 0.053$. Is $g''(h^*)$ positive or negative, and from which side is the fixed point approached?

Solution. $g'' = \beta^2 g(1-g)(1-2g)$ with $g = 0.053 < 0.5$, so $g'' > 0$. Then $\tau\dot{\delta r} = \frac{1}{2}J^2 g''\delta r^2 > 0$ on both sides: r increases. From below ($\delta r < 0$) it moves up towards r^*; from above it moves away, up towards the high state. Half-stable, approached from the left.

SECTION 12

Cheat sheet

WEEK 02 ON ONE PAGE

- **Autapse**: a neuron's synapse onto itself, strength J. $\tau\dot{r} = -r + g(Jr + h^{\text{ext}})$. Excitatory: $J > 0$ (positive feedback).
- **Linear** $g(h) = h$: $\tau\dot{r} = -(1-J)r + h^{\text{ext}}$. Leak $-r$ vs. feedback $+Jr$.
- **Fixed point**: $\dot{r} = 0$. Linear autapse: $r^* = h^{\text{ext}}/(1-J)$ ($J \neq 1$). For $0 < J < 1$, an amplification by $1/(1-J)$.
- **Stable**: nearby states return; **unstable**: nearby states run away. Only stable fixed points are seen in practice.
- **Perturbation**: $\delta r = r - r^*$; $\tau\dot{\delta r} = -(1-J)\delta r$; $\delta r(t) = \delta r(0)e^{-(1-J)t/\tau}$. h^{ext} drops out.
- $J < 1$: stable, decay time $\tau/(1-J)$ (slower as $J \to 1$). $J > 1$: unstable, divergence. $J = 1$: border (no fixed point, or a whole line of them).
- **Full solution**: $r(t) = r^* + (r(0) - r^*)e^{-(1-J)t/\tau}$.
- **Vector field**: plot $\dot{r}$ vs. r; crossing zero going down = stable, going up = unstable.
- Stability = sign of the eigenvalue of the perturbation dynamics (here $-(1-J)/\tau$), as in Week 1.

THE NON-LINEAR AUTAPSE ON ONE PAGE

- **Logistic** $g(x) = 1/(1+e^{-\beta(x-\theta)})$: between 0 and 1, $g(\theta) = 0.5$, $g' = \beta g(1-g)$, maximal slope $\beta/4$. $\beta \to \infty$: a step.
- **Fixed points**: $r^* = g(Jr^* + h^{\text{ext}})$, no closed form. Intersect the identity line with the S-curve $g(Jr + h^{\text{ext}})$ (effective gain βJ, midpoint $(\theta - h^{\text{ext}})/J$; raising h^{ext} slides it left).
- **Stability**: $\lambda = Jg'(Jr^* + h^{\text{ext}})$ = slope of the S-curve at the crossing. $\lambda < 1$ stable, $\lambda > 1$ unstable; $\delta r(t) = \delta r(0)e^{-(1-\lambda)t/\tau}$. Linear g: $\lambda = J$.
- **Global** stability: reached from every start. **Local**: reached from nearby starts. Linearisation only tests local stability.
- **Bifurcation**: qualitative change from a smooth parameter change. **Saddle-node**: a stable and an unstable fixed point merge and vanish; the curves are tangent ($\lambda = 1$).

- Book values $\theta = 0$, $\beta = 20$, $J = 1$: three fixed points for $-0.803 < h^{\text{ext}} < -0.197$, otherwise one.
- **Attractor**: region all nearby trajectories go to. **Bistable**: two stable fixed points, an unstable one between them as the watershed.
- **Memory**: a brief pulse of h^{ext} past a bifurcation point flips the state, which persists after the pulse (the state depends on history).
- **Ingredients**: positive feedback ($\beta J/4 > 1$) and saturation.
- $\lambda = 1$: second order, $\tau\dot{\delta r} = \frac{1}{2}J^2 g''\delta r^2$: **half-stable**, hyperbolic (slow) approach.